Multipretext-Task Prototypes Guided Dynamic Contrastive Learning Network for Few-Shot Remote Sensing Scene Classification

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***Abstract*— As a content management technique, remote sensing (RS) scene classification (RSSC) always attracts researchers' attention. In the past decades, many successful methods have been proposed. Nevertheless, their prerequisite is that there are large labeled datasets, which is a strict demand in practice. To resolve this contradiction, developing RSSC models with the help of few-shot learning (FSL) has become popular. Due to lacking prior knowledge, most of the existing few-shot RSSC models pay attention to the learning algorithm. However, they do not attach importance to the complex contents within RS scenes and the intricate interclass/intraclass relations between RS scenes. This would influence their performance negatively. In this article, we propose a new few-shot RSSC model named multipretext-task prototypes guided dynamic contrastive learning network (MPCL-Net). MPCL-Net consists of a multipretext tasks generation submodule, a deep feature learning submodule, and a joint optimization submodule. First, two RS-oriented pretext tasks are constructed under the self-supervised learning (SSL) framework in the multipretext tasks generation submodule, which aims to explore multiscale and rotation-invariant information from RS scenes. Second, a simple convolutional neural network (CNN) is developed in the deep feature learning submodule to transform the RS scenes into visual features. Third, three loss functions are formulated and integrated into the joint optimization submodule. Their goals are to fully capture the diverse land covers within RS scenes and compact/separate the intraclass/interclass samples with limited supervision. Finally, our MPCL-Net can be trained in a meta way. The positive results counted on the three public RS scene datasets confirm that our MPCL-Net is helpful to RSSC tasks under the few-shot scenario. Our source codes are available at https://github.com/TangXu-Group/Remote-Sensing-Images-Classification/tree/main/MPCL.**

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***Index Terms*— Dynamic contrastive learning, few-shot remote sensing scene classification (RSSC), multipretext tasks self-supervised learning (SSL).**

I. Introduction

REMOTE sensing (RS) scene classification (RSSC) is an important research topic in the RS community, which aims to assign semantic labels to RS scenes according to their contents [1], [2]. As a fundamental technique, RSSC is widely used in various RS applications, such as change detection [3], [4], contents mining [5], [6], geospatial object detection [7], [8], and so on. With the rapid development of deep learning, many RSSC methods have been proposed based on convolutional neural networks (CNNs) [9], and their performance is impressive. However, those notable results rely on a large number of labeled data. This strict requirement limits their application to reality as the volume and covered categories of labeled data are most of the time few and incomplete in practical applications. Therefore, it is necessary and urgent to develop RSSC models which can work on the few-shot scenario. In other words, the expected RSSC models should have two merits. One is that a few labeled samples can train them, and the other is that they can distinguish the new concepts.

To meet the above-mentioned demands, few-shot learning (FSL) [10] has been developed in the RS community. As a popular solution to FSL tasks, meta-learning [11] attracts researchers' attention due to its "learning to learn" mechanism. Especially, in the current deep era, many deep meta-learning algorithms have been proposed, and they have achieved success in various FSL tasks [12]. The existing meta-learning methods can be divided into three categories, including optimization-, model-, and metric-based technologies [13]. Optimization-based approaches [14], [15] aim at developing specific optimization algorithms to make models learn rapidly for new tasks. Model-based schemes [16], [17] focus on learning their internal dynamics, which can be leveraged to process and predict the new inputs. Metric-based ways [18], [19] pay attention to finding a meta-feature space and adopting the similarity comparisons to recognize the new tasks. In the RS community, most few-shot RSSC methods [20], [21], [22] are derived from metric-based meta-learning methods due to their simple and clear meta-mechanisms.

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Although those methods have reached acceptable performance and generalization, some challenges in RSSC are still to be solved. The satisfactory RSSC results cannot only depend on the similarity measures. The effectiveness and sufficient visual features for describing RS scenes are also paramount [23]. Since RS scenes have diverse and abundant land covers with multiscales, learning the useful visual features is a tough task, especially in the few-shot scenario. A natural way to address this problem is to expand the samples. However, annotating RS scenes is a time-consuming and laborious task. In contrast to the labeled data, collecting the unlabeled samples is easy. The rich information hidden in the big unlabeled data is beneficial for learning valuable features. Thus, developing the method to explore the knowledge from RS scenes without the assistance of manual annotation becomes a critical issue for the above-mentioned challenges. To this end, self-supervised learning (SSL) [24] draws researchers' attention, which aims to use the data itself to be the supervised information for completing the learning missions. A usual way of SSL for image processing is pretext task learning, which pushes the models to solve the predefined image-level tasks so that the helpful information of images can be captured. The common pretext tasks [25] are image jigsaw puzzles, image rotation, and so on. Another popular SSL way to image understanding is contrastive learning [26], which makes models distinguish the images that have a high interclass similarity or intraclass variance.

One problem is raised: is it possible to integrate the metric-based meta-learning and SSL to develop an expected RSSC model? To achieve this goal, we propose a new model named multipretext-task prototypes guided dynamic contrastive learning network (MPCL-Net) in this article. The main body of MPCL-Net consists of a multipretext tasks generation submodule and a deep feature learning submodule. The former aims to explore the multiscale and rotation-invariant information by constructing various pretext tasks under the SSL framework. The latter focuses on mapping the RS scenes into valuable visual features by a simple CNN. To train MPCL-Net using a few labeled samples, we add a joint optimization submodule at the top of MPCL-Net. This submodule involves three specific loss functions, including a multipretext-task SSL (MPT-SSL) loss, a multiview consistency FSL (MVC-FSL) loss, and a prototype-based dynamic contrastive learning (P-DCL) loss. By minimizing them jointly, our MPCL-Net can be trained with small samples. Also, its generalization can be guaranteed.

In summary, our main contributions are as follows.

1) An end-to-end RS scene classification model (MPCL-Net) is proposed with the consideration of the characteristics of RS scenes. MPCL-Net can be trained with small labeled RS samples and be used to distinguish the new RS concepts.
2) To deeply explore the complex contents within RS scenes, we introduce the SSL technique into our MPCL-Net. Along with two RS-oriented pretext tasks and a simple CNN, the valuable features that contain the multiscale and rotation-invariant knowledge can be mined from RS scenes.
3) To make the proposed MPCL-Net meet the demands of the few-shot scenario, three specific loss functions are formulated based on the conventional meta loss. Their function is to help MPCL-Net capture elegant information from RS scenes, maintain prediction consistency, and address the interclass/intraclass problem. By optimizing them jointly in a meta-manner, our MPCL-Net can be trained using a few labeled samples.
4) Extensive experiments conducted on three public RS scene datasets show that the proposed model is helpful to RS scene classification. Compared with many existing methods, the performance and generation of MPCL-Net are competitive.

The remaining sections are organized as follows. Related work is reviewed in Section II. Then, the proposed MPCL-Net is introduced in Section III. The evaluations of the proposed method are given in Section IV, Finally, the conclusions are summarized in Section V.

II. Related Work

A. Deep Remote Sensing Scene Classification Model

With the help of CNNs, a large number of RSSC models have been proposed in recent years. This section divides them into two groups for a short review. In the first group, the methods focus on improving the classification performance by reducing the influence of intraclass variance and interclass similarity in RS scenes. For example, Cheng et al. [27] proposed a discriminative CNN (D-CNN) to complete the RSSC tasks. Apart from normal cross-entropy loss, the metric-learning loss is developed in D-CNN to ensure the samples from the same (different) classes can be close to (far away from) each other. The positive results illustrate the superiorities of D-CNN. Similarly, another metric-learning-based deep model was proposed in [28]. Based on the Siamese network, Chopra et al. [29] design a metric learning regularization layer to enlarge or decrease the distances between RS scenes from the same or different classes. The reported results confirm its usefulness. Li et al. [30] first use the attention map technique to obtain the salient parts of RS scenes. Then, a CNN model with a new loss function is developed to further mine discriminative information and reduce the negative impacts of interclass dispersion and intraclass variance. Tang et al. [31] developed an attention-consistent network (ACNet) for RSSC. First, ACNet leverages the channelwise and spatialwise attention mechanisms to capture the salient regions within RS scenes. Then, the diverse salient regions are unified by an attention-consistent scheme to obtain the normalized knowledge for RSSC tasks. After the above-mentioned operations, not only can the key information hidden in RS scenes be mined, but the problem of interclass/intraclass can also be partly solved.

The approaches in the second group aim at exploring the multiscale and rotation-invariant information from RS scenes so that the models' performance in RSSC can be improved. Liu et al. [32] proposed a dual-branch multiscale CNN to capture the multiscale knowledge fully. It consists of a fixed-scale net and a varied-scale net. By forcing the outputs

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of two nets close, the obtained visual features can describe the scale information exactly. Thus, the RSSC results generated by those features are adequate. To explore the information comprehensively, a rotation invariance CNN was introduced in [33]. It combines the specific data augmentation strategy and rotation-invariant regularization to ensure the discrimination of the final visual features. Consequently, the RSSC results can be guaranteed. Lu et al. [34] introduced SIFT [35] into CNNs and designed a bidirectional adaptive feature fusion strategy to obtain the discriminative representation for RSSC. Since fused features contain both semantic (global) and scale (local) information, the classification results are as good as expected. A multiscale staking attention pooling was introduced in [36] to highlight the latent semantics and exploit the multiscale information. Through embedding the residual attention strategy and multiscale dilated convolution layer into a CNN, the effectiveness of the learned features and the accuracy of the classification results can be ensured.

B. Meta-Learning in Remote Sensing

To deal with the few-shot problem and enhance the models' generalization for diverse practical applications, meta-learning has become popular in the RS community. Based on the relational network [37], Gao et al. [38] proposed a new deep model for classifying the hyperspectral image (HSI) under the few-shot scenario. The authors first leverage the feature learning module to capture the spatial–spectral information from HSIs fully. Then, a task-based learning algorithm is designed under the meta-learning paradigm to ensure that model can classify the new HSIs. A meta-learning-based RS object detection model was developed in [39], consisting of a meta-feature extractor, a feature reweighting module, and a bounding box prediction module. This model can not only complete the detection task but also distinguish the unseen objects in the RS images. Tang et al. [5] introduced the key ideas of the prototype network [40] into the hash learning and proposed a meta-hashing model for content-based RS image retrieval. The improved metric-based meta-learning strategy helps the model learn the valuable hash codes from RS images using few labeled data.

For RSSC, meta-learning-based methods are also abundant. For instance, Li et al. [20] proposed discriminative learning of adaptive match network (DLA-MatchNet) under the meta-learning framework. With the assistance of attention mechanisms and the adaptive distance matcher, DLA-MatchNet can learn the discriminative features using a few labeled samples to accomplish RSSC tasks. Another meta-learning-based model was proposed in [21], which completes the feature learning at the task level and achieves the classification results using the Balance loss function. In [22], a Siamese-prototype network (SPNet) was developed to address few-shot RSSC tasks. Considering the properties of RS scenes, SPNet updates the normal prototype generation schemes. Along with the specific loss functions, the RSSC results are encouraging. Cui et al. [41] embedded the kernel techniques into the meta-learning framework and constructed meta-kernel networks (MKNs) for RSSC. By exploring the data distribution, the discriminative classification bias can be obtained with limited supervised information. A multiscale multiangle network and the corresponding class-level prototype guided learning algorithm were proposed in [42] for RSSC. They aim to capture the complex land covers within RS scenes based on limited labels. The reported results confirm their helpfulness. An aerial image few-shot dataset (AIFS-DATASET) was proposed in [43]. AIFS-DATASET consists of various datasets so that its data distribution is diverse enough to evaluate FSL algorithms comprehensively. At the same time, the authors also developed a dual-constrained distance metric learning (DC-DML) model in this article to deal with the FSL tasks by narrowing down and enlarging the distances between the intraclass and interclass samples.

C. Self-Supervised Learning in Remote Sensing

SSL study in RS is energetic as it can learn valuable information from RS images without supervision knowledge, which can provide auxiliary messages for many RS applications. For example, Dong et al. [44] designed a self-supervised strategy for RS image change detection, in which the developed network is asked to complete the temporal prediction. In this way, the model can mine the distribution-aware information from temporal RS images, which is beneficial to change detection. A self-supervised spectral–spatial residual network (SSRN) was proposed for HSIs super-resolution [45]. The authors first transform the super-resolution problem into a pixelwise spectral mapping issue. Then, a self-supervised fine-tuning scheme is developed to produce high spatial resolution HSIs. Another SSL model was presented for RS image semantic segmentation [46]. The encouraging segmentation results can be obtained by developing the appropriate contrastive learning modules for global styles and local regions.

In the RSSC tasks, SSL is also a prevailing technique. Li et al. [46] developed a self-supervised contrastive learning-based metric learning network (SCL-MLNet) to accomplish the few-shot RSSC tasks. First, SCL-MLNet incorporates contrastive and FSL in a multitask learning framework to obtain the representative features. Then, the attention method is adopted to separate the samples from different classes. The satisfactory classification results can be produced based on the above-mentioned two stages. In [47], SSL is utilized to generate a high-performance pretraining model for RSSC. This operation can mitigate the burden of collecting labeled data and improve classification performance. Considering the complex contents within RS scenes and the limited supervision information, Zhao et al. [48] leveraged the SSL to deal with RSSC in a multitasking way. A new mixup loss strategy is developed to weight different tasks dynamically so that the discriminative features can be learned from RS scenes at a small cost. The promising classification results confirm the idea's usefulness.

D. Similar Work Comparison

Our MPCL-Net aims to enhance the classification results by combining FSL and SSL. Admittedly, some similar solutions have been proposed for image classification, such as

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ProtoNetRot [49], SupBYOL [50], conditional self-supervised (CSS) [51], and calibrated pretext tasks (CPT) [52]. Specifically, ProtoNetRot weaves a rotation pretext task (RPT) into ProtoNet [40], where the model is required to predict the input images' rotation angles and complete FSL simultaneously. Besides the rotation prediction, SupBYOL also introduces another SSL task (representation prediction [53]) in the meta-learning stage. This way, valuable features can be learned from images under the FSL cases. Unlike the above-mentioned two methods that complete SSL and FSL simultaneously, CSS first accomplishes SSL (rotation forecast) and FSL separately. Then, the knowledge learned from two tasks is fused by a graph convolutional network (GCN) [54] module for the final classification. CPT was developed for RS images, which adds pretext task techniques and contrastive learning into the FSL framework to ensure the final performance.

Although the targets of our MPCL-Net and the four methods discussed earlier are similar, i.e., using SSL to improve FSL, there are some differences between MPCL-Net and them. For the first three methods (ProtoNetRot, SupBYOL, and CSS), the different points are as follows. First, they are general-purpose methods that are applied to natural images, while our model is developed for RS images. Apart from the common RPT, MPCL-Net also develops the scale pretext task (SPT) to mine the multiscale information hidden in RS images, which benefits RS scene classification. Second, the introduced SSL technologies in the above three methods are pretext-task-based, aiming to learn the profitable features from images by predicting the pseudolabels. Compared with them, besides developing proper pretext tasks (RPT and SPT), we also introduce contrastive learning based on feature-level prototypes into our model, which focuses on compacting the intraclass samples and pushing away the interclass samples. Third, compared with SupBYOL and CSS, the proposed MPCL-Net can be trained without pretraining, which is beneficial to practical scenarios.

For RS-oriented CPT, there are still distinct differences. First, CPT trains the base model in a supervised learning way. In other words, it leverages the whole samples in the training set directly to train the model along with a multiclass classifier by reducing the cross-entropy values. In contrast to CPT, our MPCL-Net trains the base model in an N-way-K-shot manner. The details of the N-way-K-shot learning will be discussed in Section III-A. Second, CPT only contains RPT, and MPCL-Net adopts RPT and SPT to guarantee the effectiveness of the learned features. Third, CPT uses contrast learning at the image level, and MPCL-Net applies contrast learning at the prototype level.

III. Methodology

A. Preliminary Knowledge

1) Meta-Learning: The data partition is a crucial step in meta-learning [12], [13]. Suppose there is a dataset D with a label space $\mathcal{C}$, and the number of classes is M. In general, the dataset would be divided into a training set D_{train}, a validation set D_{val}, and a testing set D_{test} under the meta-learning framework. Their corresponding label subspaces are disjoined and complementary, i.e., $C_{\text{train}} \cap C_{\text{val}} = \emptyset$, $C_{\text{train}} \cap C_{\text{test}} = \emptyset$, $C_{\text{val}} \cap C_{\text{test}} = \varnothing$, and $C{=}C_{\text{train}} \cup C_{\text{val}} \cup C_{\text{test}}$. Furthermore, the number of classes in different sets satisfies $M = M_{\text{train}} + M_{\text{val}} + M_{\text{test}}$. Meta-learning techniques train the model on D_{train}, evaluate the performance of the model on D_{val}, and assess the model's generalization to unseen data of D_{test}.

In the training stage, suppose D_{train} contains M_{train} classes with L samples per class. Since L is small, the meta-learning algorithms use a set of tasks $\mathcal{T}$ instead of single samples to train the deep models. This training paradigm is known as episode training. Each task T consists of a support set $D_s = \{(\mathrm{x}_i^s, \mathrm{y}_i^s)\}_{i=1}^{N\times K}$ and a query set $D_q = \{(\mathrm{x}_j^q, \mathrm{y}_j^q)\}_{j=1}^{N\times Q}$, which are randomly selected from D_{train}. Here, $N(N < M_{\text{train}})$ indicates the class numbers within support and query sets, K and Q are the number of samples within per class and $K + Q = L$. Then, the obtained N-way-K-shot task set $\mathcal{T}$ is used to train the deep models using the following equation:

$$\theta = \arg\min_{\theta} \sum_{\mathcal{T}} \mathcal{L}(f_\theta(T)) \tag{1}$$

where $f_\theta(\cdot)$ indicates the deep mapping of nonlinear models and $\mathcal{L}(\cdot)$ denotes the loss function that usually measures the prediction error on query sets. In the validation and testing stage, the samples within D_{val} and D_{test} should also be organized as tasks to complete the performance and generalization assessment.

2) Metric-Based Baseline Model: Our MPCL-Net is developed based on the prototypical network (ProtoNet) [40]. Thus, we introduce ProtoNet's main idea in short. ProtoNet is a simple yet classical metric-based meta-learning technique. It leverages the distances between samples within the query set and the prototypes to train the model under the few-shot scenario. In detail, for a task T, ProtoNet first computes the class-level prototypes using D_{train} according to

$$\mathbf{c}_n = \frac{1}{K} \sum_{((\mathrm{x}_i^s, \mathrm{y}_i^s)\in D_s^n)} \mathbf{f}_i^s \tag{2}$$

where $\mathbf{c}_n$ means the nth prototype, D_s^n indicates a set of samples with class n, $\mathbf{f}_i^s$ denotes the embedding of $\mathrm{x}_i^s \in D_s^n$. Then, for a query sample x_j^p, its class distribution can be calculated by

$$p\left(\mathrm{y}_j^q = n \middle| \mathrm{x}_j^q\right) = \frac{\exp\left(-d\left(\mathbf{f}_j^q, \mathbf{c}_n\right)\right)}{\sum_{n=1}^{N} \exp\left(-d\left(\mathbf{f}_j^q, \mathbf{c}_n\right)\right)} \tag{3}$$

where $\mathbf{f}_j^q$ indicates the embedding of x_j^q, and $d(\cdot,\cdot)$ denotes the Euclidean distance. By minimizing $-\log p(\mathrm{y}_j^q = n|\mathrm{x}_j^q)$, the parameters of the model can be updated by the task T. When we expand the above-mentioned process to task set $\mathcal{T}$, the model can be trained. In the reference stage, the class of a new sample would also be decided by the distances between the new sample and the corresponding prototypes.

B. Structure of the Proposed Network

The structure of the proposed MPCL-Net is shown in Fig. 1, which consists of a multipretext tasks generation submodule, a deep feature learning submodule, and a joint optimization submodule. The multipretext tasks generation

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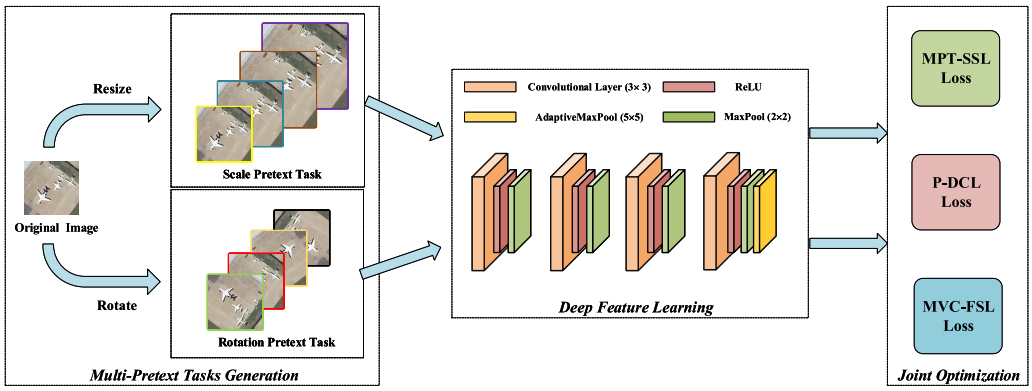


Fig. 1. Framework of the proposed MPCL-Net.

submodule focuses on preprocessing the input RS scenes and defining the RS-oriented prepretext tasks. The deep feature learning submodule aims to extract valuable information from RS scenes. The joint optimization submodule considers formulating the specific loss functions to train our MPCL-Net under the few-shot scenario. For clarity, we use a specific task T as the example to explain each submodule of MPCL-Net in the following.

The multipretext task generation submodule constructs two pretext tasks, including SPT and RPT. In SPT, the multiscale information hidden in RS scenes would be explored by randomly varying their sizes. Suppose we have a set of scale operators $G_\tau = \{g_\tau | \tau = 1, \ldots, n_\tau\}$. For an input RS scene with a size of $u \times u$, we can use g_τ to resize it into $v \times v$, where v equals $u + \tau \cdot m$ or $u - \tau \cdot m$, and m denotes the scaling margin. In RPT, the rotation-invariant messages within the input RS scenes would be captured by the simple rotation. Assume that there is a set of rotation operators $G_r = \{g_r | r = 1, \ldots, n_r\}$. We can use g_r to rotate an input RS scene with $(r-1)\cdot 90$ degrees in the 2-D space. By constructing SPT and RPT, an original input task $T = \{D_s, D_q\}$ can be transformed into two new tasks

$$T^\tau = \left\{\left\{D_s^\tau, D_q^\tau\right\} | \tau = 1, \ldots, n_\tau\right\}$$
$$T^r = \left\{\left\{D_s^r, D_q^r\right\} | r = 1, \ldots, n_r\right\}$$

where

$$D_s^\tau = \left\{\left(x_i^s, y_i^s, \tau\right)\right\}_{i=1}^{N\times K}, \quad D_q^\tau = \left\{\left(x_j^q, y_j^q, \tau\right)\right\}_{j=1}^{N\times Q}$$
$$D_s^r = \left\{\left(x_i^s, y_i^s, r\right)\right\}_{i=1}^{N\times K}, \quad D_q^r = \left\{\left(x_j^q, y_j^q, r\right)\right\}_{j=1}^{N\times Q}.$$

Like [40], our deep feature learning submodule is a simple network that consists of four convolutional blocks. Each block contains a 3×3 convolutional layer, a rectified linear unit (ReLU) layer, and a 2×2 MaxPool layer. Besides, there is an additional 5×5 adaptive MaxPool layer in the last block. After passing through the deep feature learning submodule, the RS scenes within T^τ and T^r can be mapped into the d-dimensional high-level visual features

$$\mathbf{F}^\tau = \left\{\left\{\mathbf{F}_s^\tau, \mathbf{F}_q^\tau\right\} | \tau = 1, \ldots, n_\tau\right\}$$
$$\mathbf{F}^r = \left\{\left\{\mathbf{F}_s^r, \mathbf{F}_q^r\right\} | r = 1, \ldots, n_r\right\}$$

where

$$\mathbf{F}_s^\tau = \left\{\left(\mathbf{f}_i^s\right)^\tau\right\}_{i=1}^{N\times K}, \mathbf{F}_q^\tau = \left\{\left(\mathbf{f}_j^q\right)^\tau\right\}_{j=1}^{N\times Q}$$
$$\mathbf{F}_s^r = \left\{\left(\mathbf{f}_i^s\right)^r\right\}_{i=1}^{N\times K}, \mathbf{F}_q^r = \left\{\left(\mathbf{f}_j^q\right)^r\right\}_{j=1}^{N\times Q}.$$

When we obtain these representations, the joint optimization submodule will be utilized to update the parameters of MPCL-Net. Three loss functions are developed to assist the optimization process. The first one is the MPT-SSL loss, which aims at deeply mining the valuable information from RS scenes without any manual supervision. The second one is the MVC-FSL loss. The functions of the MVC-FSL loss are twofold. One is to ensure that our MPCL-Net can be trained under the few-shot scenario. The other is to further capture the knowledge hidden in RS scenes from various views. The last one is the P-DCL loss, which focuses on addressing the problems of interclass similarity and intraclass variance using a few labeled samples. Now, we discuss three loss functions in detail.

C. Learning Algorithm for the Proposed Network

To obtain a model with good performance and high generalization using a few labeled samples, we develop the learning algorithm under the meta-learning paradigm. Here, we also use a specific task $T = \{D_s, D_q\}$ to clarify the proposed learning algorithm. The whole training process would be completed using a set of tasks $\mathcal{T}$. As mentioned in Section III-B, when the task T is passed through the multipretext tasks generation and deep feature learning submodules, we can get two pretext tasks T^τ and T^r, and the corresponding RS scenes can be mapped into the high-level features $\mathbf{F}^\tau$ and $\mathbf{F}^r$. Next, three loss functions are developed to guide the learning process. Since the loss functions of SPT and RPT are similar, we only define the different functions for SPT here for clarity. The definitions of the three loss functions for RPT can be found in the Supplementary Material.

1) MPT-SSL Loss: This loss function is developed to help MPCL-Net mine the profitable information from RS scenes without manual labels. To this end, the self-supervised scale and rotation labels $\{\tau_i\}_{i=1}^{n_\tau}$ and $\{r_i\}_{i=1}^{n_r}$ are used. For convenience, we combine the support and query sets and redefine

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$\mathbf{F}^\tau$ and $\mathbf{F}^r$ as

$$\mathbf{F}^\tau = \left\{ \{\mathbf{f}_i^\tau\}_{i=1}^{N(K+Q)} | \tau = 1, \ldots, n_\tau \right\}$$
$$\mathbf{F}^r = \left\{ \{\mathbf{f}_i^r\}_{i=1}^{N(K+Q)} | r = 1, \ldots, n_r \right\}.$$

Then, a scale loss and a rotation loss are formulated.

For T^τ, we develop a scale classifier θ_τ, consisting of a 1×1 convolutional layer and an ReLU activation function. The scale classifier realizes the mapping $\theta_\tau : \mathbf{f}_i^\tau \mapsto \tau_i$ for $(\mathbf{f}_i^\tau, \tau_i) \in \mathbf{F}^\tau$. To recognize the diverse scale variations, the scale loss is formulated as follows for the feature pair $(\mathbf{f}_i^\tau, \tau_i)$:

$$\mathcal{L}_{\text{ssl}}^\tau = -\frac{1}{n_\tau l} \sum_{\tau=1}^{n_\tau} \sum_{(\mathbf{f}_i^\tau, \tau_i) \in \mathbf{F}^\tau} \log \frac{\exp\left(\theta_\tau\left(\mathbf{f}_i^\tau, \tau_i\right)\right)}{\sum_{\tau'=1}^{n_\tau} \exp\left(\theta_\tau\left(\mathbf{f}_i^\tau, \tau_i'\right)\right)} \tag{4}$$

where $l = N(K+Q)$. For T^r, we construct a rotation classifier θ_r, whose structure is the same as θ_τ. The rotation classifier completes the mapping $\theta_r : \mathbf{f}_i^r \mapsto r_i$ for $(\mathbf{f}_i^r, r_i) \in \mathbf{F}^r$. For the feature pair $(\mathbf{f}_i^r, r_i)$, similar to Eq. (4), the rotation loss $\mathcal{L}_{\text{ssl}}^r$ can be defined to ensure the various rotations can be distinguished. Finally, the MPT-SSL loss is the combination of the scale loss and rotation loss

$$\mathcal{L}_{\text{ssl}}^{\text{mpt}} = \mathcal{L}_{\text{ssl}}^\tau + \mathcal{L}_{\text{ssl}}^r. \tag{5}$$

2) MVC-FSL Loss: This loss function has the FSL term and the MVC term. The FSL term guarantees that our MPCL-Net can be trained with limited labeled samples. The MVC term ensures that MPCL-Net can consistently predict the pretexted samples corresponding to the same RS scene.

For the RS scenes in

$$T^\tau = \left\{ \left\{ \left(x_i^s, y_i^s, \tau\right), \left(x_j^q, y_j^q, \tau\right) \right\} | \tau = 1, \ldots, n_\tau \right\}.$$

We have already obtained their features

$$\mathbf{F}^\tau = \left\{ \left\{ \mathbf{F}_s^\tau, \mathbf{F}_q^\tau \right\} | \tau = 1, \ldots, n_\tau \right\}$$
$$= \left\{ \left\{ \left(\mathbf{f}_i^s\right)^\tau, \left(\mathbf{f}_j^q\right)^\tau \right\} | \tau = 1, \ldots, n_\tau \right\}.$$

By the deep feature learning submodule, where $i \in [1, N \times K]$, $j \in [1, N \times Q]$, N denotes the class number in T^τ, K and Q indicate the numbers of the support and query samples, respectively. Similar to [40], the FSL term is formulated as

$$\mathcal{L}_{\text{fsl}}^\tau = -\frac{1}{n_\tau} \sum_{\tau=1}^{n_\tau} \log p^\tau\left(y_j^q = n \middle| x_j^q, \tau\right)$$
$$p^\tau\left(y_j^q = n \middle| x_j^q, \tau\right) = \frac{\exp\left(-d\left(\left(\mathbf{f}_j^q\right)^\tau, \mathbf{c}_n^\tau\right)\right)}{\sum_{n=1}^{N} \exp\left(-d\left(\left(\mathbf{f}_j^q\right)^\tau, \mathbf{c}_n^\tau\right)\right)}$$
$$\mathbf{c}_n^\tau = \frac{1}{K} \sum_{\left(\mathbf{f}_i^s\right)^\tau \in \mathbf{F}_{s^n}^\tau} \left(\mathbf{f}_i^s\right)^\tau \tag{6}$$

where $\mathbf{F}_{s^n}^\tau$ indicates a set of features with class n, and $\mathbf{c}_n^\tau$ denotes the prototype corresponding to the nth class. Inspired by [55], we leverage the Kullback–Leibler (KL) divergence to ensure prediction consistency. Suppose we have gotten the posterior probability distributions $\{\mathbf{p}_j^\tau | \tau = 1, \ldots, n_\tau\}$ for query samples $\{(\mathbf{f}_j^q)^\tau | \tau = 1, \ldots, n_\tau\}$ according to (6). The MVC term is defined as

$$\mathcal{L}_{\text{con}}^\tau = \frac{1}{n_\tau N Q} \cdot \sum_{\tau=1}^{n_\tau} \sum_{j=1}^{NQ} \text{mean}\left(\mathbf{p}_j^\tau \left(\log \mathbf{p}_j^\tau - \log \hat{\mathbf{p}}_j^\tau\right)\right)$$
$$\hat{\mathbf{p}}_j^\tau = \frac{1}{n_\tau} \sum_{\tau=1}^{n_\tau} \mathbf{p}_j^\tau \tag{7}$$

where mean$(\cdot)$ denotes the elemental average operation, and $\hat{\mathbf{p}}_j^\tau$ implies the center prediction distributions of the pretexted samples from one RS scene. Therefore, the MVC-FSL loss function for T^τ can be formulated as

$$\mathcal{L}_{\text{mvcfsl}}^\tau = \mathcal{L}_{\text{fsl}}^\tau + \eta^\tau \mathcal{L}_{\text{con}}^\tau \tag{8}$$

where η^τ is a free parameter to control the contributions of $\mathcal{L}_{\text{fsl}}^\tau$ and $\mathcal{L}_{\text{con}}^\tau$.

Similarly, for the RS samples in

$$T^r = \left\{ \left\{ \left(x_i^s, y_i^s, r\right), \left(x_j^q, y_j^q, r\right) \right\} | r = 1, \ldots, n_r \right\}$$

and their features

$$\mathbf{F}^r = \left\{ \left\{ \mathbf{F}_s^r, \mathbf{F}_q^r \right\} | r = 1, \ldots, n_r \right\}$$
$$= \left\{ \left\{ \left(\mathbf{f}_i^s\right)^r, \left(\mathbf{f}_j^q\right)^r \right\} | r = 1, \ldots, n_r \right\}.$$

The same process is done. Thus, the MVC-FSL loss function for T^r can be formulated as

$$\mathcal{L}_{\text{mvcfsl}}^r = \mathcal{L}_{\text{fsl}}^r + \eta^r \mathcal{L}_{\text{con}}^r \tag{9}$$

where $\mathcal{L}_{\text{fsl}}^r$ and $\mathcal{L}_{\text{con}}^r$ are the FSL and MVC terms for the rotational task, and η^r is a hyperparameter to balance the efforts of two terms.

Consequently, the MVC-FSL loss function is

$$\mathcal{L}_{\text{fsl}}^{\text{mvc}} = \frac{1}{2}\left(\mathcal{L}_{\text{mvcfsl}}^\tau + \mathcal{L}_{\text{mvcfsl}}^r\right)$$
$$= \frac{1}{2}\left(\mathcal{L}_{\text{fsl}}^\tau + \mathcal{L}_{\text{fsl}}^r\right) + \frac{\eta^\tau}{2}\mathcal{L}_{\text{con}}^\tau + \frac{\eta^r}{2}\mathcal{L}_{\text{con}}^r$$
$$\sim \underbrace{\frac{1}{2}\left(\mathcal{L}_{\text{fsl}}^\tau + \mathcal{L}_{\text{fsl}}^r\right)}_{\text{FSL Term}} + \underbrace{\eta\left(\mathcal{L}_{\text{con}}^\tau + \mathcal{L}_{\text{con}}^r\right)}_{\text{MVC Term}} \tag{10}$$

where η is an empirical parameter for the FSL and MVC terms, and $\sim$ means the equivalence.

3) P-DCL Loss: To compact/separate the intraclass and interclass samples, we develop the P-DCL loss in this section with the help of contrastive learning. A critical step in contrastive learning is constructing the positive and negative data according to the anchor sample [26]. Such that the samples corresponding to the same/different classes can be clustered/split. We apply these critical ideas to the prototypes and develop the P-DCL loss. Compared with the regular contrastive learning networks (e.g., [56], [57]), which collect the anchors and their positive and negative data pairs in the preprocessing stage at the image level, the anchors and corresponding data pairs in our P-DCL loss are the immediate class-level prototypes (generated during the training) and their variants. Therefore, the P-DCL loss can be regarded as a dynamic technique.

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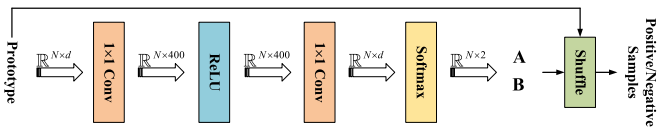


Fig. 2. Architecture of the learnable sample generator.

For T^τ, suppose we have obtained N prototypes $\mathbf{C}_N^\tau = \{\mathbf{c}_n^\tau | \tau = 1, \ldots, n_\tau\}$. First, we regard each prototype as the anchor and construct its positive and negative samples by a learnable sample generator (LSG), whose architecture is shown in Fig. 2. LSG consists of two 1×1 convolutional layers, one ReLU layer, and a Softmax layer. The inputs of LSG are the prototypes $\mathbf{C}_N^\tau$, and the outputs are two hyperparameters sets $\mathbf{A}_N^\tau$ and $\mathbf{B}_N^\tau$. After disturbing and the linear combination operations, we can get the positive and negative samples. The above-mentioned process can be formulated as

$$\begin{aligned} \mathbf{C}_N^{\tau^+} &= \mathbf{A}_N^\tau \cdot \mathbf{C}_N^\tau + \mathbf{B}_N^\tau \cdot \text{Shuffle}\left(\mathbf{C}_N^\tau\right) \\ \mathbf{C}_N^{\tau^-} &= \mathbf{B}_N^\tau \cdot \mathbf{C}_N^\tau + \mathbf{A}_N^\tau \cdot \text{Shuffle}\left(\mathbf{C}_N^\tau\right) \\ \left(\mathbf{A}_N^\tau, \mathbf{B}_N^\tau\right) &= f_{\text{lsg}}\left(\mathbf{C}_N^\tau\right) \end{aligned} \tag{11}$$

where $f_{\text{lsg}}(\cdot)$ denotes the nonlinear mapping of LSG, Shuffle$(\cdot)$ means the shuffle operation, and $\mathbf{A}_N^\tau = \{\alpha_n^\tau\}_{n=1}^N$ and $\mathbf{B}_N^\tau = \{\beta_n^\tau\}_{n=1}^N$ are hyperparameter sets. Here, α_n^τ and β_n^τ should obey the rules: $\alpha_n^\tau + \beta_n^\tau = 1$ and $\alpha_n^\tau > \beta_n^\tau$. Then, derived from InfoNCE [58], the P-DCL loss for T^τ is defined as

$$\mathcal{L}_{\text{dcl}}^\tau = -\frac{1}{n_\tau} \cdot \sum_{\tau=1}^{n_\tau} \sum_{n=1}^{N} \log \left(\frac{\exp\left(\frac{\left\langle \mathbf{c}_n^\tau, \mathbf{c}_n^{\tau^+} \right\rangle}{\varepsilon}\right)}{\exp\left(\frac{\left\langle \mathbf{c}_n^\tau, \mathbf{c}_n^{\tau^+} \right\rangle}{\varepsilon}\right) + \sum_{n'=1}^{N} \exp\left(\frac{\left\langle \mathbf{c}_n^\tau, \mathbf{c}_{n'}^{\tau^-} \right\rangle}{\varepsilon}\right)} \right) \tag{12}$$

where ε is a temperature hyperparameter and $\langle \cdot, \cdot \rangle$ implies the inner product. The target of $\mathcal{L}_{\text{dcl}}^\tau$ is to narrow down the distance between the anchor and its positive sample and enlarge the gap between the anchor and its negative sample.

To further explain how to produce the positive/negative samples, a schematic illustration of positive sample generation under the five-way-K-shot scenario for the SPTs is provided in Fig. 3. Suppose there are five prototypes $\mathbf{C}_5^\tau = \{\mathbf{c}_1^\tau, \mathbf{c}_2^\tau, \mathbf{c}_3^\tau, \mathbf{c}_4^\tau, \mathbf{c}_5^\tau\}$. After passing them through LSG, we can get two sets of hyperparameters $\mathbf{A}_5^\tau = \{\alpha_1^\tau, \alpha_2^\tau, \alpha_3^\tau, \alpha_4^\tau, \alpha_5^\tau\}$ and $\mathbf{B}_5^\tau = \{\beta_1^\tau, \beta_2^\tau, \beta_3^\tau, \beta_4^\tau, \beta_5^\tau\}$. Then, the original prototypes $\mathbf{C}_5^\tau$ and its Shuffle version would be combined linearly with the weights of $\mathbf{A}_5^\tau$ and $\mathbf{B}_5^\tau$ to generate $\mathbf{C}_5^{\tau^+}$. Since we define $\alpha_i^\tau > \beta_i^\tau$ and $\alpha_i^\tau + \beta_i^\tau = 1$, $\mathbf{C}_5^\tau$ has a higher weight in $\mathbf{C}_5^{\tau^+}$. Therefore, $\mathbf{C}_5^{\tau^+}$ can be regarded as the positive samples of $\mathbf{C}_5^\tau$.

Similarly, when we can get the positive and negative samples $\mathbf{C}_N^{r^+}$ and $\mathbf{C}_N^{r^-}$ of the prototypes $\mathbf{C}_N^r = \{\mathbf{c}_n^r | r = 1, \ldots, n_r\}$, we can also get the P-DCL loss $\mathcal{L}_{\text{dcl}}^r$ for T^r whose definition can be deduced by (12). Finally, the ultimate P-DCL loss is formulated as

$$\mathcal{L}_{\text{dcl}}^p = \mathcal{L}_{\text{dcl}}^\tau + \mathcal{L}_{\text{dcl}}^r. \tag{13}$$

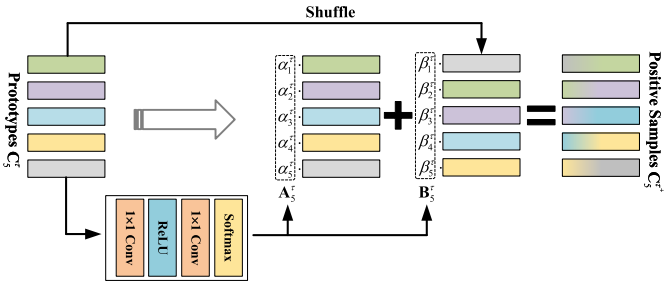


Fig. 3. Schematic illustration of positive sample generation under the five-way-K-shot scenario for the SPTs.

Algorithm 1 Steps of Training MPCL-Net

Input:

A labeled dataset that contains M classes with L samples per class, the maximum iterations *iterMax*.

1: for $i = 1$ to *iterMax* do;
2: Construct one training task T;
3: Generate scale pretext task T^τ and rotation pretext task T^r using T according to the scheme mentioned in Section III-B;
4: Pass the multi-pretext tasks through the deep feature learning submodule to get their $d-$dimensional features $\mathbf{F}^\tau$ and $\mathbf{F}^r$.
5: Compute multi-view prototypes $\mathbf{C}_N^\tau$ and $\mathbf{C}_N^r$;
6: Calculate $\mathcal{L}_{ssl}^{mpt}$, $\mathcal{L}_{ssl}^{mpt}$ and $\mathcal{L}_{ssl}^{mpt}$ by Eqs. 5, 10, and 13;
7: Calculate the final loss $\mathcal{L}_{total}$ by Eq. 14 and update the MPCL-Net's parameters by the back propagation algorithm.
8: **end for**.

Output:

The trained MPCL-Net.

4) Total Loss: Based on the contents discussed earlier, the loss function for training our MPCL-Net is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{fsl}}^{\text{mvc}} + \lambda_1 \mathcal{L}_{\text{ssl}}^{\text{mpt}} + \lambda_2 \mathcal{L}_{\text{dcl}}^p \tag{14}$$

where λ_1 and λ_2 are two hyperparameters to control the significance of different terms. Besides, we summary the training steps in Algorithm 1.

IV. Experiments

A. Datasets Description

UC Merced Land-Use (UCM)[1] [59] contains 2100 256×256 RS scenes images belonging to 21 semantic categories. Each category has 100 RS scenes. The spatial resolution of scenes is one foot. The semantics of each class and the number of scenes within each class are exhibited in Fig. 4. In our experiments, the dataset is divided into ten, six, and five classes for training, validation, and testing, respectively, and the exact splits are shown in Table I.

AID[2] [60] consists of 31 500 600×600 RS scenes, which are grouped into 30 classes. The scene numbers of each class

[1]http://vision.ucmerced.edu/datasets/landuse.html
[2]https://captain-whu.github.io/AID/

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Fig. 4. Examples of scenes of the UCM dataset. The scene numbers and names are summarized as follows. 1) Agricultural. 2) Airplane. 3) Baseball diamond. 4) Beach. 5) Buildings. 6) Chaparral. 7) Dense residential. 8) Forest. 9) Freeway. 10) Golf course. 11) Harbor. 12) Intersection. 13) Medium-density residential. 14) Mobile home park. 15) Overpass. 16) Parking lot. 17) River. 18) Runway. 19) Sparse residential. 20) Storage tanks. 21) Tennis courts.

are varied from 200 to 400. The pixel resolution of each image ranges from 8 to 1.5 m. The semantics of different classes and the number of scenes corresponding to each class are exhibited in Fig. 5. For the aerial image dataset (AID), 16 classes are randomly chosen as the training set, seven classes are selected as the validation set, and the remaining seven classes are regarded as the test set (see Table I for detail).

NWPU-RESISC45[3] [61] has 31 500 256 × 256 RS scenes, and they are divided into 45 semantic classed equally. Their spatial resolution varies from 30 to 0.2 m. The examples are exhibited in Fig. 6. In the following experiments, 25 classes are selected as the training set, ten classes are regarded as the validation set, and the rest of the ten classes are regarded as the test set, as shown in Table I.

B. Implementation Details

Following the most commonly used meta-learning protocol, we perform five-way one-shot and five-way five-shot classification. We implement our model in the PyTorch platform[4] and run our experiment on two Amax GeForce RTX 3090 GPUs. In the meta-training stage, the adaptive momentum estimation algorithm (ADAM) is used to optimize the parameters of our model. The initial learning rate is set as 1×10^{-4} and the maximum iterations are set to be 4000. We train the model from scratch without pretrained weights for each dataset. All the input images are resized to 224 × 224. In the meta-testing phase, both SPT and RPT are used to output the classification results for testing images to improve the performance with the help of multiscale and rotation-invariant information.

Several parameters should be set in advance, including the number of the scale and rotation operators n_τ and n_r, the scaling margin m, the empirical parameter η in the MVC-FSL loss [see (10)], the temperature parameter ε in the P-DCL loss [see (12)], and the hyperparameter λ_1 and λ_2 in total loss [see (14)]. The optimal values of λ_1, λ_2, and m for different datasets and diverse few-shot scenarios are exhibited in Table II, and their influence on our model will be discussed in Section IV-F. For η, as suggested by [55] and [62], we set its value to 0.1. Its impact on our MPCL-Net will also be analyzed in Section IV-F. Besides, we set $n_\tau = n_r = 4$ and $\varepsilon = 0.1$ for the following experiments. The reasons why we fix their values are twofold. First, since the valid rotation angles in RPT are $\{0°, 90°, 180°, 270°\}$, the value of n_r is decided to be four. Spontaneously, n_τ is equal to four. Second, similar to [58] and [56], we set the temperature parameter ε to be 0.1.

[3]https://gcheng-nwpu.github.io/#Datasets
[4]https://pytorch.org/



Fig. 5. Examples of scenes of the AID dataset. The scene numbers and names are summarized as follows. 1) Airport. 2) Bare land. 3) Baseball field. 4) Beach. 5) Bridge. 6) Center. 7) Church. 8) Commercial. 9) Dense residential. 10) Desert. 11) Farmland. 12) Forest. 13) Industrial. 14) Meadow. 15) Medium residential. 16) Mountain. 17) Park. 18) Parking. 19) Playground. 20) Pond. 21) Port. 22) Railway station. 23) Resort. 24) River. 25) School. 26) Sparse residential. 27) Square. 28) Stadium. 29) Storage tanks. 30) Viaduct.



Fig. 6. Examples of scenes of the NWPU-RESISC45 dataset. The scene numbers and names are summarized as follows. 1) Airplane. 2) Airport. 3) Baseball diamond. 4) Basketball court. 5) Beach. 6) Bridge. 7) Chaparral. 8) Church. 9) Circular farmland. 10) Cloud. 11) Commercial area. 12) Dense residential. 13) Desert. 14) Forest. 15) Freeway. 16) Golf course. 17) Ground track field. 18) Harbor. 19) Industrial area. 20) Intersection. 21) Island. 22) Lake. 23) Meadow. 24) Medium residential. 25) Mobile home park. 26) Mountain. 27) Overpass. 28) Palace. 29) Parking lot. 30) Railway. 31) Railway station. 32) Rectangular farmland. 33) River. 34) Roundabout. 35) Runway. 36) Sea ice. 37) Ship. 38) Snowberg. 39) Sparse residential. 40) Stadium. 41) Storage tank. 42) Tennis court. 43) Terrace. 44) Thermal power station. 45) Wetland.

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TABLE I
DATA SPLITS ON THREE DATASETS

Data Sets	Training	Validation	Testing
UCM	Agricultural; Baseball Diamond; Buildings; Chaparral; Dense Residential; Freeway; Harbor; Medium Residential; Overpass; Parking Lot;	Airplane; Forest; Intersection; Runway; Storage Tanks;	Beach; Golf Course; Mobile Home Park; River; Sparse Residential; Tennis Court
AID	Bare Land; Baseball Field; Bridge; Church; Commercial; Dense Residential; Farm Land; Forest; Industria; Medium Residential; Mountain; Parking; Pond; Port; Resort; School;	Beach; Center; Meadow; Park; Playground; Square; Stadium;	Airport; Desert; Railway Station; River; Sparse Residential, Storage Tanks; Viaduct
NWPU-RESISC45	Airplane; Baseball Diamond; Beach; Bridge; Chaparral; Church; Cloud; Desert; Freeway; Golf Course; Harbor; Island; Lake; Meadow; Mobile Home Park; Mountain; Palace; Railway; Rectangular Farmland; Roundabout; Sea Ice; Ship; Sparse Residential; Stadium; Wetland;	Commercial Area; Industrial Area; Overpass; Railway Station; Runway; Snowberg; Storage Tank; Tennis Court; Terrace; Thermal Power Station;	Airport; Basketball Court; Circular Farmland; Dense Residential; Forest; Ground Track Field; Intersection; Medium Residential; Parking Lot; River

TABLE II
EMPIRICALLY OPTIMAL VALUES OF m, AND λ_1 AND λ_2 FOR DIFFERENT DATASETS AND DIFFERENT TRAINING TASKS

	UCM			AID			NWPU-RESISC45		
	m	λ_1	λ_2	m	λ_1	λ_2	m	λ_1	λ_2
5-way 1-shot	-40	0.5	0.5	-30	0.1	0.3	-30	0.9	0.5
5-way 5-shot	20	0.5	0.4	20	0.2	0.3	-10	0.5	0.8

We adopt overall accuracy (OA) as our assessment criteria to evaluate the proposed method. OA is defined as the number of correctly classified samples divided by the total number of scenes in the dataset, and it reflects the overall classification performance.

C. Performance of Our Model

To validate our MPCL-Net extensively, we compare it with the following ten models, including matching network (MatchingNet) [63], relation network (RelationNet) [37], ProtoNet [40], ProtoNetRot [49], SupBYOL [50], DLA-MatchNet [20], SPNet [22], SCL-MLNet [46], CPT [52], and DC-DML [43]. The first five models are general-purpose FSL networks, and the last five are RS-oriented FSL networks. Their publication information can be found in Table III. Note that all models are accomplished by ourselves using the released codes, and their free parameters are set according to the original literature. In addition, for the sake of fairness, the experimental settings of the compared methods are the same as the contents discussed in Section IV-B. There is another point we want to touch on: CPT trains the base model in a supervised manner. This training way is different from our N-way-K-shot manner. To make a fair comparison between CPT and our MPCL-Net, we replace the supervised learning way in CPT with the N-way-K-shot training manner.

1) Results of the UCM Dataset: The classification performance of MPCL-Net and the compared methods counted on the UCM dataset is summarized in Table III, from where we can easily find that our MPCL-Net outperforms other models in all cases. Taking the five-way one-shot scenario as an example, compared with the other models, the improvements achieved by MPCL-Net in OAs are 8.71% (MatchingNet), 4.78% (ProtoNet), 6.52% (RelationNet), 2.36% (ProtoNetRot), 0.98% (SupBYOL), 3.83% (DLA-MatchNet), 3.62% (SPNet), 2.02% (SCL_MLNet), 0.74% (CPT), and 0.77% (DC_DML), respectively. The significant performance gaps between MPCL-Net and other methods illustrate the effectiveness of our model. Also, compared with the other methods, the standard deviations of MPCL-Net are the lowest. This implies that the feature learning ability of MPCL-Net is strong, and its behavior is stable when the volume of training data is limited. The reasons for these encouraging results can be summarized as follows. First, our model can capture complex semantic information from RS images owing to the RS-oriented pretext tasks. Second, due to the multiview consistency strategy proposed in the MVC-FSL loss, the robustness of MPCL-Net can be ensured. Last but not least, the P-DCL loss enhances the generalization of our MPCL-Net to the unseen classes.

2) Results of the AID Dataset: The OA values and standard deviations of different methods counted on the AID dataset are exhibited in Table III. Our model's classification behavior is the strongest among all the methods. For the five-way one-shot scenario, compared with the other methods, the OA improvements achieved by MPCL-Net are 12.05% (MatchingNet), 3.43% (ProtoNet), 3.02% (RelationNet), 2.29% (ProtoNetRot), 1.31% (SupBYOL), 1.83% (DLA-MatchNet), 2.59% (SPNet), 1.59% (SCL_MLNet), 0.66% (CPT), and 0.53% (DC_DML). For the five-way five-shot scenario, the OA enhancements obtained by MPCL-Net are 18.43% (MatchingNet), 5.24% (ProtoNet), 11.48% (RelationNet), 2.76% (ProtoNetRot), 1.30% (SupBYOL), 5.82% (DLA-MatchNet), 4.19% (SPNet), 4.20% (SCL_MLNet), 2.72% (CPT), and 1.61% (DC_DML). These positive results demonstrate that the proposed MPCL-Net is helpful to the AID dataset.

3) Results of the NWPU-RESISC45 Dataset: The performance of different methods counted on the NWPU-RESISC45 dataset is shown in Table III. MPCL-Net achieves the best performance in the five-way five-shot scenario. In comparison with other methods, the improvements achieved by MPCL-Net are 30.33% (MatchingNet), 2.13% (ProtoNet), 5.10% (Rela-

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TABLE III
OA (%) OF MPCL-NET AND OTHER METHODS ON THREE DATASETS

Method	Year	5-way					
		UCM		AID		NWPU-RESISC45	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
MatchingNet [63]	NeuralPS 2016	47.75±0.45	59.43±0.43	48.56±0.59	58.35±0.22	36.35±0.28	45.91±0.28
ProtoNet [40]	NeuralPS 2017	51.68±0.30	73.20±0.10	57.18±0.68	71.54±0.29	52.13±0.53	74.11±0.09
RelationNet [37]	CVPR 2018	49.94±0.34	61.51±0.24	57.59±0.24	65.30±0.17	57.70±0.36	71.14±0.12
ProtoNetRot [49]	ICCV 2019	54.10±0.65	73.78±0.57	58.32±0.49	74.02±0.36	53.44±0.23	74.78±0.18
SupBYOL [50]	arXiv 2021	55.48±0.13	75.50±0.19	59.30±0.15	75.48±0.13	54.50±0.12	75.84±0.17
DLA-MatchNet [20]	TGRS 2021	52.63±0.04	66.22±0.32	58.78±0.70	70.96±0.03	**58.76±0.18**	75.86±0.10
SPNet [22]	TGRS 2022	52.84±0.25	74.89±0.18	58.02±0.20	72.59±0.16	54.28±0.14	74.39±0.16
SCL_MLNet [46]	TGRS 2022	54.44±0.23	73.10±0.14	59.02±0.15	72.58±0.47	55.24±0.12	75.48±0.15
CPT [52]	TGRS 2022	55.72±0.38	74.05±0.22	59.95±0.54	74.06±0.29	55.26±0.63	75.33±0.31
DC_DML [43]	TGRS 2022	55.69±0.22	76.24±0.12	60.08±0.09	75.17±0.15	55.59±0.19	76.18±0.08
MPCL-Net	Ours	**56.46±0.21**	**76.57±0.07**	**60.61±0.43**	**76.78±0.08**	55.94±0.04	**76.24±0.12**

tionNet), 1.46% (ProtoNetRot), 0.40% (SupBYOL), 0.38% (DLA-MatchNet), 1.85% (SPNet), 0.76% (SCL_MLNet), 0.91% (CPT), and 0.06% (DC_DML). However, MPCL-Net cannot reach what we expect in the five-way one-shot scenario, whose behavior is weaker than that of DLA-MatchNet and RelationNet. This is because DLA-MatchNet and RelationNet adopt nonlinear classifiers, which would benefit the classification tasks when the number of classes is large, and the supervision information is limited. Similar observations can be found in [21] and [46]. Nevertheless, embedding the nonlinear classifier would increase the model's complexity. Thus, we still prefer adopting the linear classifier into our model. One of our future works is to improve the linear classifier-based model's classification performance in this kind of difficult classification scenario.

Apart from the numerical results discussed earlier, we study MPCL-Net's performance by visualizing the learned features using the t-SNE [64] algorithm. Here, we use the ski-learn library to execute the t-SNE algorithm. The function name is "TSNE," and the parameters are "n_components = 2," "init = pca," and "random_state = 0," where "n_components" indicates the dimension of the embedded space, "init" implies the initialization manner of embedding, and "random_state" decides the random number generator. Apart from the above-mentioned three parameters, other input parameters for the TSNE function equal the default values. The feature distributions of the testing data under the five-way one-shot and five-way five-shot scenarios are exhibited in Figs. 7 and 8, respectively. Due to the experimental settings (five-way), only five classes are selected from the testing sets to complete the feature distribution exhibition. For UCM, the selected five classes are "Beach," "Golf Course," "Mobile Home Park," "River," and "Tennis Courts," and they are numbered 1–5 in Figs. 7 and 8. For AID, the five classes are "Desert," "Railway Station," "Sparse Residential," "Storage Tanks," and "Viaduct." Their numbers in Figs. 7 and 8 are 6–10. For NWPU-RESISC45, the five classes contain "Airport," "Basketball Court," "Circular Farmland," "Dense Residential," and "Forest," and their numbers are 11–15 in Figs. 7 and 8. Here, the distributions of features learned by ProtoNet are also displayed for reference.

By observing Figs. 7 and 8, we can find that the discrimination of the features learned by our MPCL-Net is higher than that of ProtoNet in all cases. Taking the AID dataset and five-way five-shot setting as an example, the features obtained by ProtoNet are mixed, which cannot provide valuable information for scene classification. While the clusters of the features generated by our model are separable, the distances between different clusters are relatively apparent. Therefore, they can support discriminative clues for scene classification tasks. These positive results confirm the usefulness of our MPCL-Net.

D. Ablation Study of Loss Function

As said in Section III, the proposed MPCL-Net is developed based on ProtoNet, and it consists of a multipretext tasks generation submodule (SPT and RPT), a deep feature learning submodule (ProtoNet), and a joint optimization submodule (MPT-SSL, MVC-FSL, and P-DCL losses). To study their contributions to MPCL-Net, we construct the following networks.

1) *Net-0:* ProtoNet (with FSL loss).
2) *Net-1:* RPT + ProtoNet (with RPT-SSL and FSL losses).
3) *Net-2:* SPT + ProtoNet (with SPT-SSL and FSL losses).
4) *Net-3:* RPT + SPT + ProtoNet (with MPT-SSL and FSL losses).
5) *Net-4:* RPT + SPT + ProtoNet (with MPT-SSL and MVC-FSL losses).
6) *Net-5:* RPT + SPT + ProtoNet (with MPT-SSL, MVC-FSL, and P-DCL losses).

Net-0 is the original ProtoNet, i.e., the deep feature learning submodule. Net-1 and Net-2 are the combinations of the deep feature learning submodule and the rotation-/scale-pretext tasks generation submodule. Net-3 is our MPCL-Net with the MPT-SSL and MVC-FSL losses. Net-4 is our MPCL-Net.

The results of various networks counted on three datasets are shown in Table IV. Note that the experimental settings are the same as the contents mentioned in Section IV-B. From observing these results, we can find the following points. First, Net-1 and Net-2 are better than Net-0 since SPT and RPT can help the model capture the rich information from RS scenes. After combining SPT and RPT into MPT-SSL, the behavior

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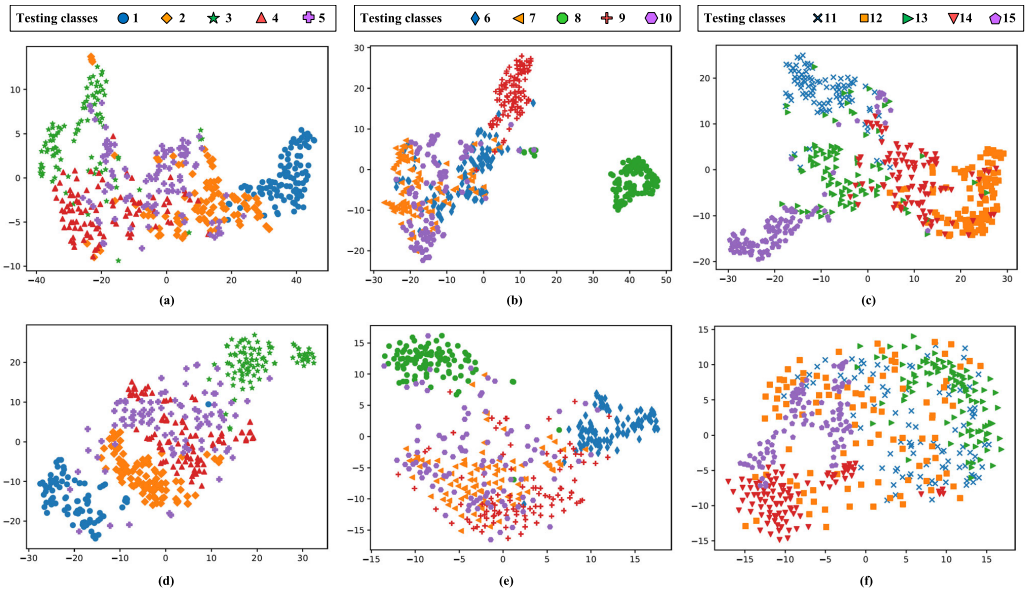


Fig. 7. t-SNE plots of tested embedding under five-way one-shot scenario. The scene numbers and names from three datasets are summarized as follows. UCM—1) Beach. 2) Golf course. 3) Mobile home park. 4) River.5) Tennis courts. AID—6) Desert. 7) Railway station. 8) Sparse residential. 9) Storage tanks. 10) Viaduct. NWPU-RESISC45—11) Airport. 12) Basketball court. 13) Circular farmland. 14) Dense residential. 15) Forest. (a) MPCL-Net on UCM. (b) MPCL-Net on AID. (c) MPCL-Net on NWPU-RESISC45. (d) ProtoNet on UCM. (e) ProtoNet on AID. (f) ProtoNet on NWPU-RESISC45.

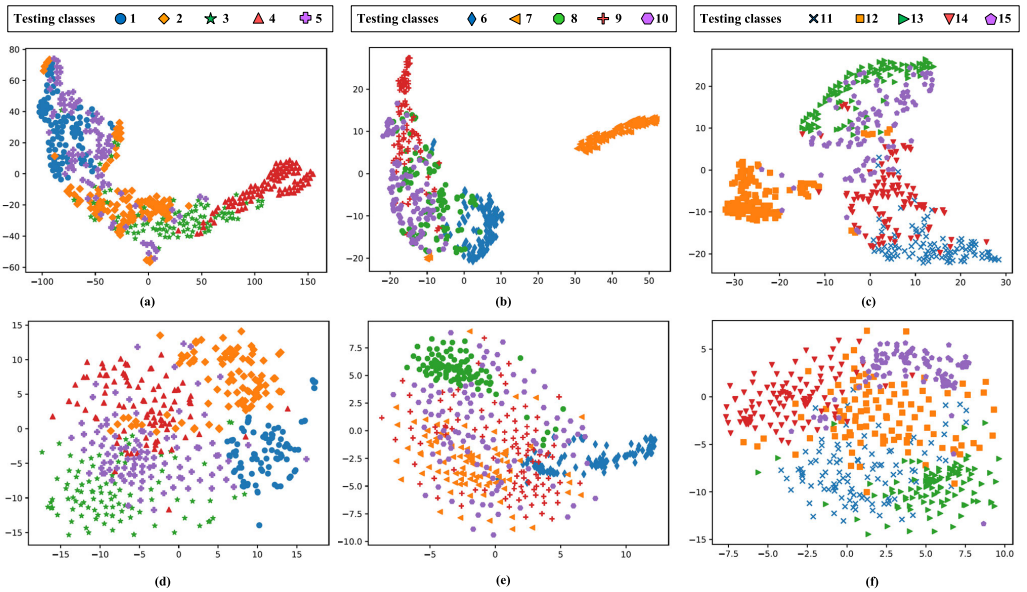


Fig. 8. t-SNE plots of tested embedding under five-way five-shot scenario. The scene numbers and names from three datasets are summarized as follows. UCM—1) Beach. 2) Golf course. 3) Mobile home park. 4) River. 5) Tennis courts. AID—6) Desert. 7) Railway station. 8) Sparse residential. 9) Storage tanks. 10) Viaduct. NWPU-RESISC45—11) Airport. 12) Basketball court. 13) Circular farmland. 14) Dense residential. 15) Forest. (a) MPCL-Net on UCM. (b) MPCL-Net on AID. (c) MPCL-Net on NWPU-RESISC45. (d) ProtoNet on UCM. (e) ProtoNet on AID. (f) ProtoNet on NWPU-RESISC45.

of Net-3 becomes stronger, which explicitly indicates the effectiveness of our multipretext tasks generation submodule and the MPT-SSL loss. Furthermore, when our MVC-FSL loss replaces the original FSL loss, Net-4 outperforms Net-3 in all cases, and its standard deviations are less than Net-3 on three datasets. This implies the positive influence of the MVC term. Adding the proposed P-DCL loss to Net-4, the classification results of Net-5 are better than that of Net-4, which shows the importance of the dynamic contrastive learning strategy in RSSC. In sum, each part of MPCL-Net plays a positive role in the final classification.

E. Statistical Test

To further evaluate the differences between the proposed MPCL-Net and the comparison methods, we adopt the Wilcoxon rank sum test [65], [66], [67], which is a non-parametric statistical test method. Particularly, all methods are randomly carried out in 30 rounds to construct the various OA pairs. Then, the Wilcoxon rank sum test (0.05 significant level) is calculated between different OA pairs. In general, the test results would be symbolized by "+," "−," and "=," where "+" and "−" indicate that there is a difference between

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TABLE IV
OA (%) OF DIFFERENT ABLATION MODELS ON THREE DATASET

Method	5-way					
	UCM		AID		NWPU-RESISC45	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Net-0	51.68±0.30	73.20±0.10	57.18±0.68	71.54±0.29	52.13±0.53	74.11±0.09
Net-1	54.03±0.53	73.81±0.67	58.13±0.60	73.44±0.34	53.37±0.18	74.80±0.21
Net-2	52.96±0.48	73.77±0.26	57.74±0.06	75.14±0.54	54.85±0.63	74.60±0.37
Net-3	54.25±0.39	74.26±0.21	58.47±0.36	75.22±0.07	54.85±0.16	75.11±0.23
Net-4	55.72±0.31	74.91±0.09	58.97±0.23	76.35±0.20	55.23±0.17	75.35±0.31
Net-5	**56.46±0.21**	**76.57±0.07**	**60.61±0.43**	**76.78±0.08**	**55.94±0.04**	**76.24±0.12**

TABLE V
WILCOXON RANK SUM TEST (0.05 SIGNIFICANCE LEVEL) OF MPCL-NET AND OTHER METHODS ON THREE DATASETS

Method	5-way					
	UCM		AID		NWPU-RESISC45	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
MatchingNet [63]	4.76e+01(1.74e-01)-	5.95e+01(1.53e-01)-	4.84e+01(1.20e-01)-	5.84e+01(9.79e-02)-	3.64e+01(7.92e-02)-	4.59e+01(1.14e-01)-
ProtoNet [40]	5.16e+01(1.63e-01)-	7.33e+01(7.43e-02)-	5.74e+01(1.09e-01)-	7.15e+01(9.29e-02)-	5.22e+01(2.15e-01)-	7.41e+01(4.73e-02)-
RelationNet [37]	5.04e+01(5.80e-01)-	6.15e+01(1.12e-01)-	5.75e+01(1.30e-01)-	6.55e+01(7.57e-02)-	5.77e+01(8.18e-02)+	7.12e+01(4.18e-02)-
ProtoNetRot [49]	5.44e+01(1.68e-01)-	7.35e+01(1.17e-01)-	5.92e+01(2.60e+00)-	7.40e+01(1.60e-01)-	5.34e+01(8.08e-02)-	7.47e+01(1.15e-01)-
SupBYOL [50]	5.54e+01(9.66e-02)-	7.55e+01(9.44e-02)-	5.93e+01(1.17e-01)-	7.54e+01(6.86e-02)-	5.45e+01(7.93e-02)-	7.59e+01(6.17e-02)-
DLA-MatchNet [20]	5.30e+01(9.87e-01)-	6.63e+01(1.74e-01)-	5.90e+01(3.52e-01)-	7.10e+01(7.43e-02)-	**5.87e+01(9.20e-02)+**	7.58e+01(7.08e-02)-
SPNet [22]	5.29e+01(8.33e-02)-	7.49e+01(1.18e-01)-	5.89e+01(1.26e-01)-	7.26e+01(9.91e-02)-	5.43e+01(5.12e-02)-	7.44e+01(4.42e-02)-
SCL_MLNet [46]	5.44e+01(1.45e-01)-	7.32e+01(5.45e-02)-	5.91e+01(1.04e-01)-	7.23e+01(7.25e-02)-	5.53e+01(6.35e-02)-	7.55e+01(7.88e-02)-
CPT [52]	5.58e+01(1.56e-01)-	7.41e+01(6.21e-02)-	5.99e+01(1.37e-01)-	7.41e+01(1.07e-01)-	5.54e+01(3.88e-01)-	7.53e+01(6.52e-02)-
DC_DML [43]	5.57e+01(9.99e-02)-	7.63e+01(7.62e-02)-	6.01e+01(7.72e-02)-	7.52e+01(6.57e-02)-	5.56e+01(5.20e-02)-	7.62e+01(6.00e-02)-
MPCL-Net (Ours)	**5.64e+01(1.19e-01)**	**7.65e+01(8.05e-02)**	**6.04e+01(9.91e-02)**	**7.68e+01(1.11e-01)**	5.60e+01(5.81e-02)	**7.63e+01(9.85e-02)**

MPCL-Net and the compared method, and "=" denotes no difference exists. Furthermore, "−" implies that the performance of MPCL-Net is better than the compared method, and "+" shows the opposite. The test results are summarized in Table V. We can easily find that there is no "=," indicating that the proposed MPCL-Net differs from the selected compared models. Besides, apart from the five-way one-shot scenario of NWPU, the MPCL-Net achieves the best performance, which is consistent with the information presented in Table III. These investigations demonstrate that the proposed method over different methods is statistically significant.

F. Sensitivity Analysis

In this section, we study the influence of empirical parameters on the proposed model, including the empirical parameters λ_1 and λ_2 [see (14)], the scaling margin m, and the hyperparameter η [see (10)]. To this end, we train a set of MPCL-Nets with different parameters. Then, their performance is counted on different datasets under the five-way one-shot scenario. In particular, the values of λ_1, λ_2, and η are varied from 0.1 to 1.0 with an interval of 0.1, and the values of m are changed from −40 to 40 with an interval of 10 (except 0). The results are shown in Fig. 9. Note that only one parameter is turned at a time, and the others are set according to the contents described in Section IV-B.

From the observation of Fig. 9(a), we can find that the optimal values of λ_1 are 0.5 (UCM), 0.1 (AID), and 0.9 (NWPU-RESISC45). For UCM, the performance of MPCL-Net increases when λ_1 is varied from 0.1 to 0.5. When $\lambda_1 > 0.5$, the behavior of our model becomes weaker but stable. For AID, the performance trend of MPCL-Net is down with the increase of λ_1. Inversely, MPCL-Net's behavior on NWPU-RESISC45 becomes stronger when λ_1 gets bigger.

By investigating Fig. 9(b), it is easy to find that the peak values of λ_2 appear at 0.5 (UCM), 0.3 (AID), and 0.5 (NWPU-RESISC45). Similar to λ_1, the performance of MPCL-Net gradually increases with time at first and then decreases. The demarcation point is $\lambda_2 = 0.5$. Contrast to λ_1, when $\lambda_2 \neq 0.3$ and $\lambda_2 \neq 0.5$ the proposed model's performance on AID and NWPU-RESISC45 is approximately equal. These discoveries suggest that the satisfactory λ_2 can be found at around 0.5.

For m [see Fig. 9(c)], its optimal values are −40 (UCM), −30 (AID), and −30 (NWPU-RESISC45). As a whole, the performance trend of MPCL-Net for three datasets is down when m increases. Also, when the values of m are positive, the behavior of various MPCL-Nets becomes steady. Those findings imply that shrinking RS scenes is beneficial to our model.

For η, by investigating the bars, we can find that the optimal performance appears when $\eta = 0.1$, and the behavior of MPCL-Nets gets weaker with the increasing η. This is because the MVC term in the MVC-FSL loss focuses on improving the model stability by the multiview consistent prediction. However, when its weight grows, it pushes the model to generate similar (or even the same) features for samples of different views. Therefore, the discrimination of the final feature representation would be reduced so that the classification results would be influenced negatively.

G. Influence of Training Samples Volume

To further study how the number of training data affects our model, we conduct the experiments under various five-way-K-

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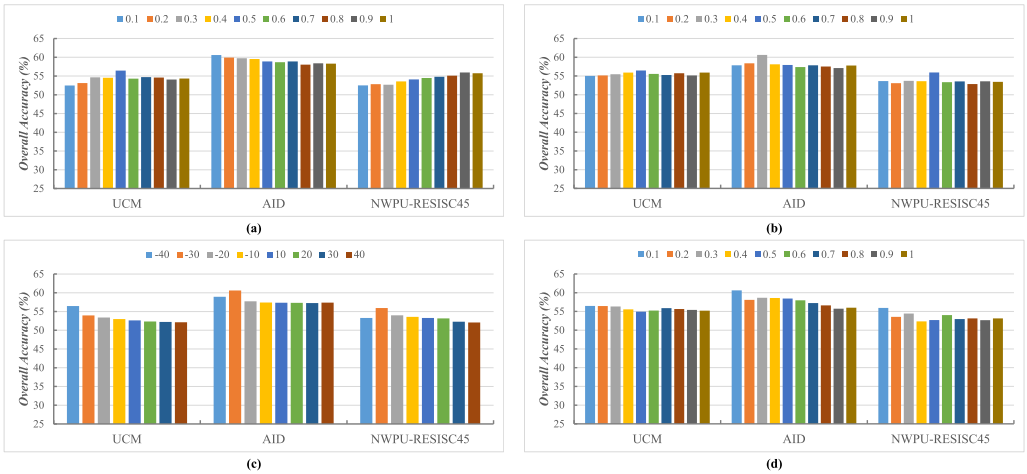


Fig. 9. Influence of empirical parameters on the proposed model. (a) λ_1. (b) λ_2. (c) m. (d) η.

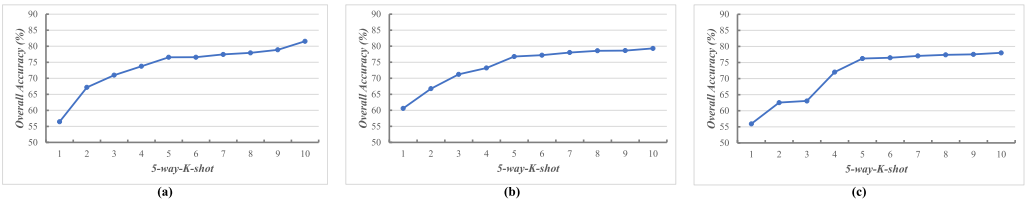


Fig. 10. Performance of MPCL-Net under various five-way K-shot scenarios counted on three datasets. (a) UCM. (b) AID. (c) NWPU.

TABLE VI

PARAMETERS (MB) AND INFERENCE TIMES (s) OF DIFFERENT ABLATION NETWORKS UNDER FIVE-WAY ONE-SHOT SCENARIO COUNTED ON THE UCM DATASET

Methods	Parameters	Inference Times
Net-0	0.113	34
Net-1	0.119	51
Net-2	0.119	51
Net-3	0.126	67
Net-4	0.126	67
Net-5 (MPCL-Net)	0.450	67

TABLE VII

PARAMETERS (MB) AND INFERENCE TIMES (s) OF MPCL-NET AND COMPARED METHODS UNDER FIVE-WAY ONE-SHOT SCENARIO COUNTED ON THE UCM DATASET

Methods	Parameters	Inference Times
MatchingNet [63]	0.112	26
ProtoNet [40]	0.113	34
RelationNet [37]	2.759	16
ProtoNetRot [49]	0.119	51
SupBYOL [50]	0.226	38
DLA-MatchNet [20]	5.091	28
SPNet [22]	0.113	34
SCL_MLNet [46]	19.159	40
CPT [52]	0.427	34
DC_DML [43]	20.552	74
MPCL-Net(ours)	0.450	67

shot scenarios by changing the number of shots from 1 to 10. The results counted on three datasets are displayed in Fig. 10. Note that the free parameters of MPCL-Net are the same as the five-way one-shot and five-way five-shot cases when $K < 5$ and $K \geq 5$. We can easily find that the behavior of our MPCL-Net becomes stronger when the number of shots increases. Furthermore, when shots rise from 1 to 5, the MPCL-Net's performance enhancements are distinct. When shots are raised from 5 to 10, the performance of MPCL-Net is improved steadily. The above-mentioned investigations confirm that the proposed MPCL-Net performs reasonably when the number of training samples changes.

H. Parameters and Inference Times

To study the efficiency of our MPCL-Net, we count its parameters and inference time under a five-way one-shot scenario on the UCM dataset. First, we count the parameters volume and inference times of each part of MPCL-Net. To this end, the networks constructed in the ablation study section (Section IV-D) are used. Their parameter sizes and inference times are summarized in Table VI. We find that the multi-pretext tasks generation submodule makes the inference times longer distinctly, and the P-DCL loss increases the model's parameters dramatically. These observations are reasonable because: 1) compared with the single pretext, preparing SPT and RPT is a time-consuming process and 2) the positive and negative sample generation procedure for the P-DCL loss would multiply the model's sizes.

Second, the compared models' parameters volume and inference times are counted for comparison (see Table VII). For the parameters, we can observe that MPCL-Net is significantly lighter than ReLationNet, DLA-MatchNet, SCL_MLNet, and DC_DML. Although our model is more

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complex than others, the parameter increments are slight. As a whole, the proposed MPCL-Net is not a heavy model. For the inference times, MPCL-Net is relatively time-consuming. This is because of the multipretext tasks generation submodule. In other words, most time is spent on constructing the pretext tasks. Fortunately, considering the positive classification performance, a longer inference procedure is not an unacceptable issue. Speeding up our model is one of our future works.

V. Conclusion

This article presents a new network for few-shot RSSC tasks named MPCL-Net. First, a multipretext tasks generation submodule is developed under the SSL framework to explore the complex contents hidden in RS scenes from the multiscale (SPT) and rotation-invariant (RPT) aspects. Then, a simple yet useful CNN is adopted to map the RS scenes into useful deep features. Finally, a specific loss function with MPT-SSL, MVC-FSL, and P-DCL terms is formulated under the metric-based meta-learning paradigm to train MPCL-Net using limited supervision. On the one hand, owing to the introduction of SSL and FSL ideas, MPCL-Net can be trained in a meta way. On the other hand, since involving the multiview scheme and the contrastive learning theory, MPCL-Net can not only cope with the interclass/intraclass issue but also distinguish the unseen classes. The encouraging experimental results on three public datasets demonstrate the effectiveness and generalization for few-shot RSSC tasks.

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